

IMO 2026 — Problem 1

Solution by GPT-5.6 Sol

2026-07-17

Source: `results/imo-2026/gpt-5.6-sol/problem-01/current.md` — autonomous run, no human intervention; solution as written by the model.

Problem

There are 2026 integers greater than 1 written on a blackboard, not necessarily different. In a move, Confucius chooses two integers $m > 1$ and $n > 1$ from different places on the blackboard and replaces these two integers with $\gcd(m, n)$ and $\frac{\text{lcm}(m, n)}{\gcd(m, n)}$. He continues to make moves while it is possible to do so. (a) Prove that, regardless of the choices of Confucius, after finitely many moves, exactly one integer M on the blackboard is greater than 1. (b) Prove that the value of M does not depend on the choices of Confucius.

Solution document

Status

solved

Approaches tried

- Arithmetic product/content invariant — worked: the strictly decreasing positive integer $2^{\mathbf{kP}}$ proves termination, and the gcd of each prime-exponent column determines the terminal nonunit.
- Valuation-vector normal form — worked: the strictly decreasing integer $\mathbf{E}+\mathbf{k}$ proves termination, and the subgroup generated by each exponent column determines the unique terminal vector.

Current best

Every possible play terminates with exactly one entry greater than 1. If the initial entries are $a_1, \dots, a_{2026}$, that terminal entry is

$$M = \prod_{p|a_1 \cdots a_{2026}} p^{\gcd(v_p(a_1), \dots, v_p(a_{2026}))},$$

so it is independent of all choices.

Full proof

Let the initial entries be $a_1, \dots, a_{2026}$. We use the **invariant/monovariant method** and **divisor analysis prime by prime**, as listed in the knowledge base.

At any stage, let P be the product of all entries and let k be the number of entries greater than 1. We claim that the positive integer

$$\Psi = 2^k P$$

strictly decreases after every legal move.

Suppose $m, n > 1$ are selected and put $g = \gcd(m, n)$. By the gcd-lcm product identity,

$$\gcd(m, n) \operatorname{lcm}(m, n) = mn.$$

The product of the two replacement entries is therefore

$$g \cdot \frac{\operatorname{lcm}(m, n)}{g} = \operatorname{lcm}(m, n) = \frac{mn}{g},$$

so the new board product is $P' = P/g$.

If $g=1$, the outputs are $1, mn$, so $k' = k-1$ and $\Psi' = \Psi/2$. If $g>1$ and $m=n$, the outputs are $m, 1$, so $k' = k-1$ and $\Psi' = \Psi/(2m)$. Finally, if $g>1$ and $m \neq n$, both outputs exceed 1: the second could equal 1 only if $\operatorname{lcm}(m, n) = g$, which would force $m=g=n$. Thus $k' = k$ and $\Psi' = \Psi/g \leq \Psi/2$. These cases are disjoint and exhaustive, and in every case Ψ strictly decreases. Hence no infinite sequence of moves exists.

A move is possible exactly when at least two entries exceed 1; consequently every terminal board has at most one such entry.

It remains to show that there is exactly one and that its value is fixed. Let

$$S = \{p : p \text{ is prime and } p \mid a_1 \cdots a_{2026}\}.$$

This finite set is nonempty because every initial entry exceeds 1. For each $p \in S$, define

$$d_p = \gcd(v_p(a_1), \dots, v_p(a_{2026})),$$

including zero valuations. Since at least one valuation in this column is positive, $d_p > 0$.

Fix a prime p . If the selected entries have valuations $r = v_p(m)$ and $s = v_p(n)$, the standard valuation formulas, consequences of the Fundamental Theorem of Arithmetic, show that the replacement valuations are

$$(\min(r, s), |r - s|).$$

These two pairs have the same gcd. Indeed, if $r \geq s$, then an integer divides both r, s exactly when it divides both $s, r-s$; the case $s \geq r$ is symmetric. This includes equality and zero valuations. Adjoining all unchanged valuations proves that the gcd of the complete p -exponent column is invariant under a move.

A terminal board cannot consist entirely of ones: for any $p \in S$, its terminal exponent-column gcd would then be 0, contradicting the invariant positive value d_p . Therefore every terminal board has exactly one entry $M > 1$, proving part (a).

At such a board, the $\mathbf{p}$ -exponent column consists of zeros except for $\mathbf{v}_{\mathbf{p}}(\mathbf{M})$. Its gcd is therefore $\mathbf{v}_{\mathbf{p}}(\mathbf{M})$, and invariance gives

$$v_p(M) = d_p \quad (p \in S).$$

No prime outside $\mathbf{S}$ can appear, because its valuations begin all zero and the update sends $(0,0)$ to $(0,0)$. The Fundamental Theorem of Arithmetic now yields

$$M = \prod_{p \in S} p^{d_p} = \prod_{p | a_1 \cdots a_{2026}} p^{\gcd(v_p(a_1), \dots, v_p(a_{2026}))}.$$

This depends only on the initial entries and not on the choices of moves, proving part (b).